DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING						
Course Code: 24CS340	*Course Title: Computer Organization*					
Total Credits: 03 *L-45: T-0: P-0: SL and TW-45*		*SDGs addressed: 4,9*	*Course Category: PCC*	*Course Type: Theory*		
Teaching Learning Process		*Total Contact Hours/semester*	*Assessment Weightage and Marks*			
	Lecture	45		CIE	SEE	Total
	Tutorial	-	*Weightage, %*	*40*	*60*	*100*
	Practical	-	*Maximum Marks*	*40*	*60*	*100*
	Self-learning and Term Work	45	*Minimum Marks*	*20*	*25*	*45**
	Total	90				

Note: *For passing the student has to score a minimum of 45 Marks (CIE+SEE: 20 + 25 or 21 + 24)

Course Prerequisite: NIL

Course Objectives:

Sl. No.	Course Objectives
1	Understand the structure, function and characteristics of computer system.
2	Identify the elements of modern instructions sets and their impact on processor design.
3	Learn different ways of communicating with I/O devices and hierarchical memory systems like cache memory and virtual memory.
4	Describe ALU operations with integer and floating–point operands and understand the basic processing unit and organization of simple processor and other large computing systems.

Course Outcomes (COs)

CO	Course Outcomes	Highest Level of Cognitive Domain	Working Knowledge
CO1	Apply the basic concepts of computer, instruction set architecture, function and characteristics of a computer system and design the assembly language for the sequence of machine instruction.	L3	WK2, WK3
CO2	Explain the different ways of communicating with the I/O devices.	L2	WK3, WK4
CO3	Design and analyse the performance of memory systems.	L3	WK2, WK3
CO4	Apply the Arithmetic and Logical operations with integer and floating–point operands.	L3	WK2, WK3
CO5	Illustrate the basic processing unit and organization of simple processor.	L2	WK3, WK4

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

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Course Content / Syllabus:

UNIT No.	Content	Lecture Hours
1	**Basic Structures of Computers, Machine Instructions & Programs:** Operational Concepts, Number Representations and Arithmetic Operations, Character Representation, Performance. **Instruction Set Architecture:** Memory Locations and Addresses, Memory Operations, Instructions and Instruction Sequencing, Addressing Modes, Assembly Language, Stacks, Subroutines, Additional Instructions, CISC Instruction Sets, RISC and CISC styles. **Self-Learning and Term Work*** • The Stack frames for nested subroutines • Additional Instructions: Multiplication and Division	09
2	**Input/Output Organization:** Basic Input/Output, Accessing I/O Devices, Interrupts, Bus Structure, Bus Operation, Arbitration. **Self-Learning and Term Work*** • An example of RISC –Style I/O Program • An example of CISC –Style I/O Program • Examples of Interrupt Programs	09
3	**The Memory System:** Basic Concepts, Semiconductor RAM Memories, Read Only Memories, Direct Memory Access, Memory Hierarchy, Cache Memories, Performance Considerations, Virtual Memory. **Self-Learning and Term Work*** • Examples of Mapping Techniques • Numericals on Performance Considerations	09
4	**Arithmetic Operations:** Addition and Subtraction of Signed Numbers, Design of Fast Adders, Multiplication of Signed Numbers, Fast Multiplication: Bit-Pair Recoding of Multipliers, Integer Division, Floating-point Numbers and Operations. **Self-Learning and Term Work*** • Examples of Floating-point operations	09
5	**Basic Processing Unit:** Some Fundamental Concepts, Instruction Execution, Hardware Components: Register File, ALU, Datapath, Hardwired Control: Generation of Control Signals, CISC style Processors **Self-Learning and Term Work*** • CISC style Processors: An interconnect using buses	09

NOTE: *Self-Learning and Term Work are evaluated only through continuous internal evaluation (CIE) and will not be assessed during the Semester End Examination (SEE).

Text Book:

Sl. No.	Author/s	Title	Publisher Details
1	Carl V Hamacher, Zvonko Vranesic	Computer Organization and Embedded Systems	6th Edition, McGraw Hill, 2012

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Reference Books:

Sl. No.	Author/s	Title	Publisher Details
1	M. MorisMano	Computer Systems Architecture	3rd Edition, Pearson/PHI, 2007.
2	Carl Hamacher, Zvonko Vranesic, Safwat Zaky	Computer Organization	5th Edition, Tata McGraw Hill, 2002.
3	William Stallings	Computer Organization & Architecture	9th Edition, Pearson, 2015
4	V Heuring and H Jordan	Computer System Design &Architecture	1st Edition, Addison-Wesley, 1999.

Web Resources:

Sl. No.	Web Links
1	https://nptel.ac.in/courses/106106092/
2	https://www.youtube.com/watch?v=CDO28Esqmcg

Course Articulation Matrix:

Course Outcomes	Program Outcomes											Program Specific Outcomes		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	3	-	-	-	-	-	-	-	-	3	3	3
CO2	3	3	-	-	-	-	-	-	-	-	-	3	3	-
CO3	3	3	-	-	-	-	-	-	-	-	-	3	3	3
CO4	3	3	3	-	-	-	-	-	-	-	-	3	3	3
CO5	3	3	3	3	-	-	-	-	-	-	-	3	3	-

High – 3, Medium – 2, Low – 1

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